

MCRS — Modular Captured Ride System

CMFSB ISOLATION MODULE — DOCUMENT 00

Master Engineering Baseline

MCRS-MOD-001 · Architecture: CMFSB — Captured Mechanical Floating Safety Body

Always captured. Soft when you want comfort. Locked when you need control. Settles on passive support when power is gone.

Revision 5.29 · Contract Diligence Baseline · Licence and IP-Transaction Negotiation

Field	Comments
Document ID	MCRS-DOC-00-R5.29 (including overlay issues R5.25–R5.27). All are WITHDRAWN and are not design inputs.
Issue date	2025-01-15 The prefix ATFSV is retired; MCRS is the controlling identifier.
Supersedes	All overlay of a corrective addendum on the R5.25 body, and R5.27). All are WITHDRAWN and are not design inputs. The prefix ATFSV is retired; MCRS is the controlling identifier.
Document class	Module-Primary External Baseline
Classification	T1 — Confidential / NDA controlled. This file contains no T3 content. The IP schedule is a separate T3 document (MCRS-DOC-00-T3A, counsel-controlled, own revision control) and is not part of this file.
Legal Owner / Author	Amirbahador Atai (individual rights holder; assignment to a legal entity is a separate controlled act)
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Status	Contract Diligence Baseline. Not certified. TRL 2-3.
Independent Technical Reviewer	TBD — to be appointed no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit
Approval authority	TBD after independent review
Next revision triggers	ICD-001 draft · SIM-001 run · STR-001 stop curve · ACT-001 actuator map · ENE-001 envelope · packaging re-baseline decision · IP filing-status update · reviewer disposition

Controlled limitation. This document supports T1 technical diligence, supplier evaluation, Path A licence negotiation and IP-transaction discussion. It is not a performance certificate, type approval, safety case, measured ride-comfort claim, production commitment, warranty, patent grant, price list or regulatory approval. OPEN items remain OPEN until evidence closes them.

Document at a glance

Informative summary. Controlled values are in §3 and §7 only.

Topic	Position at R5.29
Product	MCRS-MOD-001: a four-tower isolation module that keeps the upper body mechanically captured at all times, with a COMFORT isolation mode, a SPORT mechanical lock and passive support on power loss.
Maturity	TRL 2-3. Evidence level L0 (architecture and requirements). No simulation, bench or vehicle result exists yet.
Offer	Path A: module licence, primarily under a paid Option (SIM-001 run and ICD-001 draft). Paths B and C are secondary. No prices appear in this document.
Principal features	Safety invariants independent of the Lorentz control loop; quantitative structures for power loss, rescue, lateral-g policy, stop curve, horizontal path and sensor independence; one-ladder demonstration and evidence plan.
Principal open items	(1) Packaging stack does not close at the quoted heights (§4). (2) Force constant Km is an assumption (§3, §10). (3) Core-tower mass limit is not met at P90 (§6.1). (4) Numeric safety limits are not yet frozen (§7). (5) No patent application is on record as filed (§16.5; the Legal Owner confirms before each release). (6) Total electrical demand is not yet budgeted (§5).
Disclosure	T1 under NDA. Subject to the T1-release precondition in Annex D.3.

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0. Revision control

0.1 Identity rule. One document ID identifies exactly one content set. Only values in §3 (Master Parameter Register) and

0.2 Revision history.

Rev	Date	Summary
R5.29	2026-10-07	Professional control issue. Supersedes R5.28. Single content layer retained. No technical parameter or requirement deleted. Identity, supersedes chain and revision-history wording unified for negotiation packs.
R5.28	2026-10-07	Single controlling baseline, one layer. Corrections to the withdrawn R5.27: revision label unified across cover, identifier and end marker; MASS-001 registered in the document register; test-protocol identifier aligned to MCRS-TP-001; patent-filing status worded consistently as “per project record, confirmed by the Legal Owner”; fault-dwell parameter registered (PARAM-FAULT-DWELL); recovery from an S1-origin S3 entry defined; at-a-glance wording neutralised.

0.3 Naming. MCRS (Modular Captured Ride System) is the product and programme name. CMFSB (Captured Mechanical Floating Safety Body) is the architecture name. MCRS-MOD-001 is the licensable module. Use of the name MCRS in external material is subject to counsel clearance (open item 17).

1. Purpose, non-claims, value, product framing and disclosure tiers

1.1 Purpose. This baseline defines the licensable CMFSB isolation module (MCRS-MOD-001) for Path A engagement and supports Path B host studies and Path C IP discussion under counsel. It fixes architecture language, safety invariants, dual-mode behaviour, controlled requirements, verification planning and disclosure tiers for T1 diligence.

1.2 Non-claims register (binding for external communication; single location).

#	Non-claim
1	No production readiness or series-production commitment
2	No type approval, certification or regulatory compliance of module or vehicle
3	No certified ASIL assignment
4	No measured ride-comfort percentage or guaranteed comfort improvement until L1/L2 configuration-managed evidence
5	No crashworthiness or survivability demonstration
6	No guaranteed compliance with the host mass ceilings (PARAM-GVW-VL, PARAM-GVW-VH) from the module alone
7	No final supplier mass, cost or lead-time commitment
8	No clinical outcome claim for ambulance / RA-1 use
9	No continuous public-road high-speed capability claim
10	No closed mass or energy budget suitable for production release
11	No patent grant, novelty, enforceability or freedom-to-operate conclusion before counsel-led analysis
12	No FIA homologation or series eligibility from FIA-referenced design philosophy
13	No supplier or OEM endorsement
14	No continuous RMS force capability until thermal bench data exist
15	No warranty, price list or offer of exclusive sale by this document
16	No packaging feasibility closure at the quoted overall heights (§4)
17	No claim that the core-tower mass limit is met at the declared confidence level (§6.1)
18	No use of the word “final” to imply package closure, verified performance, safety approval or IP protection

1.3 Value statement and reference application RA-1. Occupant compartments in specialty, defence-utility, ambulance and extreme limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both by keeping the upper body mechanically captured at all times, providing a COMFORT isolation envelope and a SPORT mechanical lock, with passive support on power loss. RA-1 (ambulance / clinical transport) is the reference use context for comfort-related requirements; no clinical outcome is claimed (non-claim 8). COMFORT addresses the vibration path; SPORT and capture address sudden attitude change under manoeuvre. This is a requirements-context statement only. EN 1789 and EN 1865 are planning references for RA-1 hosts under Path B. Path A does not require an RA-1 host.

1.4 Product framing — three tangible states (architecture, not marketing claims). The product line is: Always captured. Soft when you want comfort. Locked when you need control. Settles on passive support when power is gone. The four clauses map one-to-one onto the safety invariants and modes of §2. They may be used in T0 material and T1 executive summaries provided no measured comfort percentage, certification or production commitment is attached. The final wording of the last clause is subject to counsel review (liability, R-12).

State	Occupant / operator experience (qualitative)	Architecture behaviour	State ID
COMFORT	Road harshness is isolated from the cabin	Locks released; macro and micro stages active; leveling active	S2
SPORT	Body feels planted for rapid manoeuvres	Locks confirmed; isolation stages held	S2S
Power loss	Cabin settles on a known mechanical path	Valves fail-closed; passive support; software not required	S4

1.4.1 External teaser metrics (T0 / first contact — qualitative only). Unless L1/L2 evidence later upgrades them under configuration control, public or first-contact material uses only the following.

Teaser point	Controlled wording	Shall not be extended to
Capture	Upper body remains mechanically captured in all operating states	Levitation or free-floating claims
Dual mode	Selectable COMFORT isolation and SPORT mechanical lock	Measured handling-superiority percentages
Mode time	Mode-transition target class ≤ 3 s, excluding interlock inhibit time (PARAM-MODE-TIME)	Guaranteed production timing
Power loss	Fail-closed passive support on power loss (S4 invariant)	Unlimited duration or any-road claims

No module-mass figure is used in T0 or first-contact material. Apart from the mode-time target class above, all numeric values stay behind NDA and carry their status labels. Presenting STUDY or ASSUMPTION values as verified performance is a communication defect.

1.5 Planning KPIs (method definitions only — not measured claims).

KPI	Metric definition	Method	Reference	Status
Ride isolation	aw (0.5–80 Hz) + VDV; MSDV (0.1–0.5 Hz)	SIM-001, then bench	ISO 2631-1; ISO 8608	OPEN
Handling stiffness	Roll angle / lateral response, SPORT vs COMFORT, same step	SIM-001, mule	Defined lateral proxy	OPEN
Mode time	Accepted request to declared state (PARAM-MODE-TIME)	Bench / HIL	§7.1	TARGET
Power loss	Settle on passive support inside the §7.2 envelope	Power-cut test	S4 invariant	BASELINE
Module mass	Core tower at PARAM-MASS-CONFIDENCE; P50 and P90 both reported	Weighing	§6.1	TARGET

KPI rows marked OPEN stay OPEN until L1 evidence exists.

1.6 Disclosure tiers.

Tier	Content	Access condition
T0	Public non-enabling teaser material (§1.4 and §1.4.1 only)	Public or first contact
T1	This confidential engineering baseline and the MOD-001 datasheet derived from it	NDA
T2	Models, detailed ICD, test data, configuration-controlled design artefacts, detailed actuator map	NDA + need-to-know
T3	Patent claim charts, prior-art mapping, filing and ownership record, negotiation strategy, commercial terms (COM-001)	NDA + counsel + restricted list

MOD-001 is T0-compatible: it contains only §1.4 and §1.4.1 content, the §16.7 table and the current evidence level. Commercial conditions (Option validity, exclusivity, fees, credit) are T3 content in COM-001 only; T1 holds a one-line reference.

1.7 Status vocabulary (exclusive — one label per row; qualifiers go in a Note column).

Label	Meaning
BASELINE	Architectural decision locked for this revision; verification still required
TARGET	Numeric design goal; not verified by test
STUDY	Indicative planning figure; subject to change
ASSUMPTION	Input taken without programme-owned evidence
DERIVED	Calculated from stated inputs; inherits their uncertainty
OPEN / TBD	Unresolved at this revision
OPTION	Not committed in the baseline
LOCKED	Programme engagement decision for geometry / mass; not test verification and not packaging closure
SUPERSEDED / HISTORICAL	Prior content retained for traceability only

Standards use a separate vocabulary: **Reference** (applied as a source of rules or definitions) and **Candidate** (under consideration; compliance not claimed).

1.8 Low-complexity demonstration ladder. Buyer confidence rises faster from visible demonstration than from additional architectural options. No step is a performance claim until its evidence record exists under configuration control.

Step	Artefact	Purpose	Complexity	Status
D0	30–45 s architecture animation (layers, COMFORT, SPORT, power loss to S4)	Shared mental model	Low — visualisation only	OPTION
D1	Single-corner desk model or 3-D printed kinematic mock-up	Show capture and lock geometry	Low — non-functional	OPTION
D2	SIM-001 comparative plot pack (COMFORT vs frozen passive)	First numerical isolation evidence (L1)	Medium — analysis	TARGET
D3	Single-tower bench (force, thermal, power loss, lock cycle)	Hardware evidence (L2)	Medium-high	TARGET
D4	Four-corner / HIL mode transitions	System evidence (L3)	High	TARGET

Recommended sequence: complete D0 before wide T1 distribution; commit D2 under Option; hold D3 and D4 as milestone evidence. Gates are in §13.1.

1.9 Permitted and non-permitted claims at the current evidence level (L0).

Permitted	Not permitted without new, reviewed evidence
The architecture and requirements are defined at TRL 2–3	Production readiness
The system is intended to remain mechanically captured	Proven crashworthiness or occupant safety
COMFORT, SPORT lock and passive-support concepts are specified	Verified comfort or handling superiority
Quantitative work packages and evidence gates are identified	Validated total electrical demand
The correct external description is “Contract Diligence Baseline — TRL 2–3; open verification gates remain”	Compliance with a named safety standard; ASIL assignment
	“Patent pending” without a filing receipt; patentability, exclusivity or FTO assurance
	Describing any issue as “final” in a way that implies package closure, verified performance, safety approval or IP protection

2. Architecture, safety invariants and modes

2.1 Architecture. The upper body is kinematically constrained at all times (no free levitation). Four corner towers each combine: (1) a hydraulic macro stage (PARAM-M02-MACRO-STROKE) with leveling and lock seats; (2) a Lorentz micro stage (PARAM-M03-MICRO-STROKE) within the PARAM-LORENTZ-GAP envelope; (3) progressive mechanical stops; (4) an optional SPORT positive lock. A shared ECU implements mode arbitration, the REQ-M21 interlock and power-loss detection. Layer stack, top to bottom: upper occupant / safety body → micro stage → macro stage → lower chassis. Active isolation operates inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded. This principle remains valid if the Lorentz micro stage is replaced by another high-bandwidth actuator. **Scope:** this baseline contains no attitude, counter-lean or pyramidal underbody feature; any such feature would be a separate configuration-controlled option with its own evidence plan.

2.2 Independent load and force paths.

Path	Carrier
Normal vertical load	Passive / compliant mechanical-hydraulic support
Comfort dynamic force	Lorentz micro stage (force-bounded)
Large displacement and leveling	Hydraulic macro stage
Crash / abnormal load	Mechanical capture + progressive stops (independent of the Lorentz closed loop)
Horizontal load	Capture frame, links, horizontal stops, SPORT lock faces (§2.6)
Power loss	S4 passive support inside the §7.2 envelope

2.3 Safety invariants (non-negotiable behavioural rules).

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture
Static support	Vehicle weight on passive or fail-safe compliant paths — never the sole magnetic path
Power loss (S4)	Passive support; leveling valves fail-closed; software aliveness not required; envelope per §7.2
Crash path	Progressive stops + structure; independent of closed-loop magnetic force
Mode transition	SPORT↔COMFORT never enters unconstrained free-float
SPORT lock fault	Lock failure shall not defeat capture or progressive stops
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock (REQ-M25)
Override limit	Driver override cannot defeat the lateral-g interlock (REQ-M21)
Manual rescue	Two-step rescue (REQ-M15) remains available under power loss and hydraulic failure

2.4 Dual-mode definition. Selection inputs: driver command (highest priority), radar / surface proposal, GPS / map proposal. The default after self-test is COMFORT unless ICD-001 states otherwise. The lateral-g interlock policy is defined once, in §7.4.

Mode	State	Mechanical state	Objective
COMFORT	S2	Locks released; macro and micro active; leveling active	Soft ride isolation
SPORT	S2S	Upper body seated; all required locks confirmed; isolation stages held	Handling stiffness (roll / yaw)

2.5 Hold function (REQ-M16). Hold is an optional secondary rate-limited restraint, distinct from the SPORT positive lock and from progressive stops. Implementation class: hydraulic orifice and/or friction device at the macro stage. Engagement velocity and force limits are OPEN (frozen in ICD-001). Hold is not a primary crash-energy path. Its mass is included in the corner hydraulic / stop share. If hold is not fitted, REQ-M16 is N/A for that configuration.

2.6 Horizontal restraint. Macro travel and progressive stops are specified primarily in the vertical axis; braking, cornering and crash introduce horizontal loads. Load paths: (1) capture frame and linkage shear capacity; (2) dedicated horizontal stops or links with defined free play; (3) SPORT lock faces providing additional shear / roll restraint when engaged. REQ-M25 owns numeric free play, stiffness and load share; REQ-M26 protects the Lorentz gap. Per-class horizontal magnitudes are in §6.3.

3. Master Parameter Register

Only values in this section and requirement text in §7 are active design inputs. One status per row. ASSUMPTION, STUDY and OPEN values shall not be presented externally as verified performance.

Parameter ID	Value	Status	Note
PARAM-M02-MACRO-STROKE	±50 mm body-relative (100 mm total range)	TARGET	
PARAM-M03-MICRO-STROKE	±7.5 mm	TARGET	
PARAM-LORENTZ-GAP	2-5 mm coil-magnet baseline	TARGET	15-30 mm applies only to the PM+EM OPTION
PARAM-FORCE-PEAK	±2.5 kN per tower	STUDY	
PARAM-FORCE-RMS	≈0.7 kN RMS per tower	STUDY	Planning value
PARAM-KM	≥63 N/√W at 2-5 mm gap	ASSUMPTION	Requires FEA / supplier map (ACT-001)
PARAM-COPPER-MEAN	≤0.5 kW four towers at PARAM-FORCE-RMS	TARGET	Check: $4 \times (700/63)^2 \approx 494$ W
PARAM-COPPER-PEAK	≈1.57 kW per tower; ≈6.3 kW four towers	DERIVED	Copper loss only: $(2,500/63)^2 \approx 1,575$ W per tower
PARAM-SUPPLY-NOM	48 V class nominal	STUDY	
PARAM-BW-CONTROL	≥20 Hz control bandwidth	TARGET	Carriage first structural mode ≥3× (≥60 Hz)
PARAM-K-EFF	≈40 kN/m about the leveled operating point	STUDY	
PARAM-FN-STUDY	≈1.23 Hz at the occupied corner share	DERIVED	$\sqrt{(40,000/662.5)}/2\pi = 1.237$ Hz
PARAM-REF-MASS-OCC	≈2,650 kg occupied upper-body reference	STUDY	
PARAM-REF-MASS-MAX	≈3,150 kg maximum-study upper body	STUDY	
PARAM-CRASH-MASS-BASIS	Crash and stop sizing on PARAM-REF-MASS-MAX	BASELINE	
PARAM-OCC-POLICY	100 kg per occupant (all host budgets)	BASELINE	
PARAM-GVW-VL	4,000 kg V-L hard ceiling	LOCKED	Engagement target
PARAM-GVW-VH	7,000 kg V-H hard ceiling	LOCKED	Engagement target
PARAM-TOWER-MASS-P50	48 kg core tower	STUDY	
PARAM-MASS-CONFIDENCE	P90	BASELINE	Declared in this revision; changed only under CFG-001
PARAM-TOWER-MASS-LIMIT	≤55 kg core tower at PARAM-MASS-CONFIDENCE	TARGET	Not met by current estimate (R-01)
PARAM-TOWER-MASS-P90-LINESUM	67 kg (line-sum of element P90 values)	STUDY	Exceeds the limit by 12 kg; conservative upper bound
PARAM-TOWER-MASS-P90-RSS	≈59 kg (root-sum-square of element P90-P50 deltas added to P50)	DERIVED	Indicative only; not a verified percentile (\$6.1); exceeds the limit by ≈4 kg
PARAM-STATIC-RULE	Floor = $1.25 \times F_{max}$, rounded up to the next 0.5 kN; F_{max} = computed maximum static corner load rounded up to the next 0.25 kN	BASELINE	
PARAM-STR-S-STATIC-FLOOR	≥4.5 kN per corner	TARGET	Class S
PARAM-STR-M-STATIC-FLOOR	≥7.0 kN per corner	TARGET	Class M
PARAM-STR-L-STATIC-FLOOR	≥10.0 kN per corner	TARGET	Class L
PARAM-FREEPLAY-V	40 mm vertical free play to stop	STUDY	One value for the \$4 stack and the \$6.4 energy illustration
PARAM-CABIN-H-ROOF	≥1,000 mm from H-point to roof	TARGET	Cabin definition used in §4
PARAM-FLOOR-HPOINT	≈250 mm from cell floor to H-point (floor + seat height)	ASSUMPTION	Owner confirms in CFG-001 before G1
PARAM-M21-AY-HIGH	0.35 g (SPORT→COMFORT inhibit)	TARGET	Temporary freeze
PARAM-M21-AY-LOW	0.30 g (re-enable hysteresis)	TARGET	Temporary freeze
PARAM-M21-HOLD	Re-enable hold time	OPEN	Frozen by SIM-001 / HIL
PARAM-FAULT-DWELL	Class A diagnostic-clear dwell time (REQ-M28)	OPEN	Frozen in ICD-001 / SAF-001
PARAM-M21-ROLL-GRAD	≤6.0 deg/g COMFORT at design payload (V-L)	TARGET	
PARAM-MODE-TIME	≤3 s, SPORT↔COMFORT	TARGET	Accepted request to declared state; excludes REQ-M21 inhibit time
PARAM-LIFE	≥10,000 h equivalent road duty	TARGET	
PARAM-REVIEWER-DEADLINE	Earlier of 2026-12-05 and G1 closure	BASELINE	Process parameter

4. Packaging feasibility gate

Packaging rule. LOCKED geometry in §14 is a programme engagement target only; it is not packaging closure. Until H-point, cabin interior height, floor-to-H-point, tower integration, macro travel, free play, stop stroke, micro-stage clearance and chassis integration are configuration-controlled under CFG-001, the residual margin may be negative. The residual shall not be forced non-negative by reducing cabin height below H-point requirements. A negative residual is a P0 open item, not a hidden assumption. No negative residual may be hidden by reducing H-point, occupant clearance, required stop stroke or service access. In every external-facing table the quoted heights are read as “LOCKED TARGET / PACKAGING GATE OPEN”.

Vertical stack-up, SPORT / seated. The cabin is defined from the H-point (PARAM-CABIN-H-ROOF), so the floor-to-H-point segment (PARAM-FLOOR-HPOINT) is part of the stack.

Element	V-L (mm)	V-H (mm)	Status	Note
Overall height (SPORT)	1,700	2,000	LOCKED	Engagement target
Ground clearance (SPORT)	200	300	LOCKED	Engagement target
Available for structure	1,500	1,700	DERIVED	Height minus ground clearance
Chassis / axle structure	280	320	STUDY	Integrate tower where possible
Tower structure + mounts	120	140	STUDY	Partial integration in chassis
Macro travel (100) + free play (PARAM-FREEPLAY-V)	140	140	STUDY	Stop stroke and micro clearance not included
Floor-to-H-point	250	250	ASSUMPTION	PARAM-FLOOR-HPOINT
Cabin, H-point to roof	1,000	1,000	TARGET	PARAM-CABIN-H-ROOF
Roof / exterior shell	80	90	STUDY	
Allocated sum	1,870	1,940	DERIVED	
Residual margin (baseline report)	-370	-240	DERIVED	OPEN — P0
Residual excluding floor-to-H-point (reference only)	-120	+10	DERIVED	Not a baseline value
Stop stroke (STR-001)	not in stack	not in stack	OPEN	Further reduces both residuals
Micro-stage clearance, tolerance, thermal and sag allowance	not in stack	not in stack	OPEN	

Conclusion. At the quoted engagement heights and the current allocation the stack does not close. The module consumes at least 260 mm of vehicle height (tower 120 mm + travel and free play 140 mm) before stop stroke. Closure options, none selected: (a) integrate tower structure and travel inside chassis depth; (b) raise overall height (V-L would need about +370 mm on the current allocation); (c) reduce ground clearance; (d) package free play inside the cell underfloor. A re-baseline decision on overall height, ground clearance and cabin definition is due at G1. The residual is re-reported at every revision until ≥ 0 under CFG-001. Path A does not depend on host height; it depends on the module vertical envelope above.

Closure evidence (G1). CFG-001 freezes a datum-based 3-D stack including H-point, floor-to-H-point, cabin roof, macro stroke, free play, micro stage, stop stroke, tolerances, deflection, service envelope and chassis interfaces. G1 closure requires signed package drawings, a worst-case tolerance stack, independent review, and either a residual ≥ 0 or an approved re-baseline of the LOCKED heights.

5. Power and current envelope

5.1 Rule. PARAM-COPPER-PEAK is Lorentz copper loss at simultaneous peak force on four towers. It is not module electrical demand. Electrical input power equals copper loss plus mechanical power plus driver / converter losses. Total system power shall include: Lorentz actuators; hydraulic pump; valve actuation; lock actuation; ECU and sensors; transient and inrush; driver / converter losses; safety and thermal derating margins. ENE-001 and ACT-001 define: continuous, RMS and peak current; peak duration; simultaneous-tower demand; driver efficiency; bus voltage tolerance; cable and connector losses; fuse and protection strategy; thermal derating; transient and regenerative conditions; simultaneous events; wiring voltage drops; fuse / breaker coordination; connector ratings; EMC; fault-current limits.

Consumer	Mean (W)	Peak (W)	Status	Owner
Lorentz, four towers — mean	≤ 500	—	TARGET	ACT-001
Lorentz, four towers — peak (copper only)	—	$\approx 6,300$	DERIVED	ACT-001
Hydraulic pump	OPEN	OPEN	OPEN	HYD-001
Valves (leveling + hold)	OPEN	OPEN	OPEN	HYD-001
Lock actuators	OPEN	OPEN	OPEN	LCK-001
ECU and sensors	OPEN	OPEN	OPEN	ICD-001
Total system	OPEN	OPEN	OPEN	ENE-001

5.2 Order-of-magnitude check (DERIVED, lower bound). PARAM-COPPER-PEAK at a 48 V nominal bus corresponds to at least ≈ 131 A (≈ 33 A per tower) before mechanical power and converter losses. Wiring, driver, fusing and 48 V-class limits (candidate reference ISO 21780) are sized in ACT-001 / ENE-001 from the measured force constant and driver efficiency, not from this figure.

5.3 Claim restriction. No total peak-power or supply-adequacy claim is permitted until ENE-001 has been reviewed against supplier data and the worst-case duty cycle.

6. Mass, static floors and load cases

6.1 Module mass and statistical basis. PARAM-TOWER-MASS-LIMIT applies to the core tower only (structure + Lorentz + sensors + local electronics) and is stated at PARAM-MASS-CONFIDENCE (P90). The limit is not demonstrated: P50 is 48 kg, the P90 line-sum is 67 kg and the indicative RSS estimate is ≈ 59 kg. The mass target shall not be described as met. The line-sum is a conservative upper bound; the RSS figure is not a verified percentile because it assumes independent elements without documented distributions. The aggregate P90 is calculated by a documented Monte-Carlo or otherwise justified statistical method using supplier-supported element distributions, declared correlation assumptions, declared inclusion / exclusion boundaries and stated uncertainty. If the aggregate P90 stays above the limit, the design is revised or the requirement is formally re-baselined under CFG-001 before G2 (R-01). Core tower mass is always quoted with both P50 and P90.

Element	P50 kg	P90 kg	Scope
Local structure (per tower)	18	25	Core
Lorentz + local carriage	22	30	Core
Sensors + local electronics	8	12	Core
Core tower subtotal	48	67	Limit 55 kg at P90 — not met by current estimate
Cylinder + accumulator share	15	22	Corner
Valves / hoses / seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness / dampers / bellows	4	6	Corner
Pump + reservoir share (¼)	8	12	Shared
Full corner assembly (Class L)	94	137	STUDY; rounded to ≈ 95 kg in the host WBS
Four corners (Class L)	376	548	Rounded to ≈ 380 kg in the host WBS (§14.3)

6.1.1 Per-class corner mass (planning, STUDY).

Class	Mapped host	Corner P50 (kg)	Four-corner P50 (kg)	Note
S	V-L (4-seat)	≈ 70	≈ 280	Lighter hydraulic share
M	Intermediate	≈ 85	≈ 340	Interpolated until supplier data exist
L	V-H / heavy	≈ 95	≈ 380	Rounded from the 94 / 376 kg line sums

6.2 Static support floor — one rule, traceable inputs. Rule per PARAM-STATIC-RULE. Maximum static load = corner mass share $\times 9.81$.

Class	Mass basis	Computed static / corner	Fmax used	$1.25 \times F_{\max}$	Floor
S (V-L)	Occupied corner share 310–350 kg	3.04–3.43 kN	3.5 kN	4.375	≥ 4.5 kN
M	Corner share 450–550 kg	4.41–5.40 kN	5.5 kN	6.875	≥ 7.0 kN
L	PARAM-REF-MASS-MAX $\div 4 = 787.5$ kg	7.73 kN	7.75 kN	9.69	≥ 10.0 kN

The Class S traceability to the V-L budget is in §14.3.

6.3 Reference load cases (STUDY). All values are corner mass share $\times 9.81 \times$ multiplier. Class L is shown on both mass bases; under PARAM-CRASH-MASS-BASIS the sizing basis is the maximum-study column.

Case	Class S	Class M	Class L (2,650 kg ref)	Class L (3,150 kg max) — sizing basis
Corner mass share	310–350 kg	450–550 kg	662.5 kg	787.5 kg
Static / tower	≈ 3.0 – 3.4 kN	≈ 4.4 – 5.4 kN	≈ 6.5 kN	≈ 7.7 kN
Static floor	≥ 4.5 kN	≥ 7.0 kN	—	≥ 10.0 kN
3 g operational / tower	≈ 9 – 10 kN	≈ 13 – 16 kN	≈ 19.5 kN	≈ 23.2 kN
20 g vertical / tower	≈ 61 – 69 kN	≈ 88 – 108 kN	≈ 130 kN	≈ 154.5 kN
$\times 2$ asymmetry / tower	≈ 120 – 140 kN	≈ 175 – 215 kN	≈ 260 kN	≈ 309 kN
Horizontal 1 g, total (upper)	≈ 12 – 14 kN	≈ 18 – 22 kN	≈ 26 kN	≈ 30.9 kN
Horizontal 20 g, total (upper)	≈ 245 – 275 kN	≈ 355 – 430 kN	≈ 520 kN	≈ 618 kN

Risk R-08 compares Class L crash loads with capture-frame, stop and hardpoint capacity.

6.4 Stop-energy illustration (STUDY). $E = m \cdot a \cdot s$, equivalently $\frac{1}{2} \cdot m \cdot v^2$ with $v = \sqrt{(2 \cdot a \cdot s)}$, for $a = 20$ g and $s =$ PARAM-FREEPLAY-V (40 mm): $v \approx 3.96$ m/s; $E \approx 20.8$ kJ for 2,650 kg and ≈ 24.7 kJ for 3,150 kg (sizing basis). The illustration assumes a 20 g body-to-chassis deceleration over the free play. STR-001 replaces it with the design energy event: initiating scenario, relative closing velocity from the crash pulse and the S5 geometry, free-play geometry, corner versus whole-body share and mass case. Until the

force-displacement curve is frozen, all energy figures remain STUDY.

7. Controlled requirements (REQ-M01 to REQ-M28)

Gate = earliest gate at which objective evidence is expected. Status OPEN means a numeric limit or procedure is not yet frozen — not that the requirement is optional. Verification method: T test, A analysis, I inspection, D demonstration.

7.1 Requirement table.

ID	Requirement	Owner	Gate	Verif.	Status
REQ-M01	Upper body mechanically captured in all states; static capacity per class S/M/L at the \$6.2 floors	Structures	G0 / G2	A, T	BASELINE
REQ-M02	Macro stroke PARAM-M02-MACRO-STROKE without clash	Mechanical	G1	A, T	TARGET
REQ-M03	Micro stroke PARAM-M03-MICRO-STROKE inside PARAM-LORENTZ-GAP	Actuator	G1	A, T	TARGET
REQ-M04	Control bandwidth and carriage first mode per PARAM-BW-CONTROL	Control	G2	A, T	TARGET
REQ-M05	Peak and RMS force and Km per the register; Km stays ASSUMPTION until ACT-001	Actuator	G2	T	STUDY
REQ-M06	Core tower mass per PARAM-TOWER-MASS-LIMIT at PARAM-MASS-CONFIDENCE	Module	G2	I	TARGET
REQ-M07	Power-loss behaviour with the quantitative criteria of \$7.2	Safety	G2	T	BASELINE
REQ-M08	Multi-band SIM-001 acceptance per \$12	Simulation	G1	A	TARGET
REQ-M09a	Progressive-stop force-displacement curve with the content of \$7.5, frozen as screening curve in STR-001	Structures	G2	A	TARGET
REQ-M09b	Stops absorb the design energy without dependence on the magnet force loop (verified)	Structures	G3	T	TARGET
REQ-M10	Life per PARAM-LIFE	Reliability	G3	T, A	TARGET
REQ-M11	Operating temperature range frozen	Thermal	G2	T	OPEN
REQ-M12	EMI and cabin-field plan	EMC	G3	T	OPEN
REQ-M13	ICD mechanical / electrical / communication / safety / lock: draft released at G1, frozen at G2	Systems	G1	I	TARGET
REQ-M14	No single failure causes loss of mechanical capture	Safety	G2	A, T	TARGET
REQ-M15	Two-step manual rescue per \$7.3	HF / Safety	G3	D	TARGET
REQ-M16	If hold is fitted: rate-limited orifice / friction; not a primary crash path (\$2.5)	Safety	G2	T	BASELINE
REQ-M17	Required SPORT locks positively confirm engagement and release; lock time is an allocation inside PARAM-MODE-TIME (value in ICD-001)	Lock	G2	T	TARGET
REQ-M18	COMFORT provides the full macro envelope without lock interference	Mechanical	G2	T	TARGET
REQ-M19	Mode arbitration driver / radar / GPS with override hierarchy	Controls	G3	A, T	STUDY
REQ-M20	No transition enters unconstrained free-float	Safety	G2	A, T	BASELINE
REQ-M21	SPORT+COMFORT lateral-g policy per \$7.4 and roll gradient PARAM-M21-ROLL-GRAD	Dynamics	G2	A, T	TARGET
REQ-M22	Leveling preserves dynamic stroke under payload change (PARAM-K-EFF is the rate about the leveled point)	Hydraulics	G2	T	TARGET
REQ-M23	Incomplete lock set inhibits SPORT declaration; lock state diagnosable and reported to the ECU	Lock / Controls	G2	T	TARGET
REQ-M24	Fluid systems observe ISO 4413 safety rules	Hydraulics	G1	I, A	TARGET
REQ-M25	Horizontal load path per \$7.6, defined and verified before any integration-ready claim	Structures	G2	A, T	OPEN
REQ-M26	Horizontal relative motion allowed by REQ-M25 free play shall not reduce the Lorentz coil-magnet gap below its minimum, or the carriage shall decouple from horizontal motion	Structures / Actuator	G1	A	OPEN
REQ-M27	Sensor functional independence per \$7.7	Safety / Systems	G2	A, T	TARGET
REQ-M28	Fault recovery per the fault classes and rules of \$7.8; no automatic return to S2 or S2S from a critical, active, stale, intermittent or unclassified fault	Safety / Controls	G2	A, T	TARGET

Leveling note (REQ-M22). Static sag is taken by gas-spring preload and active leveling, not by the dynamic stroke. A payload change of 2.4 kN gives ≈ 60 mm at PARAM-K-EFF, so active leveling is required to preserve dynamic stroke.

PARAM-MODE-TIME is measured from accepted request to declared state (S2S or S2) and excludes REQ-M21 inhibit time. The lock-time allocation, the detection and valve-response times of \$7.2 and the inhibit exclusion are frozen in ICD-001 (open item 16).

7.2 REQ-M07 — Quantitative power-loss behaviour. After loss of commanded active power the system shall: (1) detect the loss within a defined maximum detection time $t_{det,max}$; (2) transition to S4; (3) close the relevant valves within a defined maximum response time $t_{valve,max}$; (4) remain inside the mechanical capture envelope; (5) remain below a defined maximum vertical displacement z_{max} and velocity v_{max} ; (6) prevent uncontrolled free-float; (7) keep passive support at or above the \$6.2 floor for the applicable class, with residual support capacity and environmental conditions stated, under loss of electrical power and the defined hydraulic failure cases. All limits are frozen in ICD-001 at G2; verification is analysis plus fault-injection and bench evidence at L2 and L3. State definition alone is not acceptance. Isolation valves are located at the cylinder port so that fail-closed action isolates a burst line (ICD design requirement).

7.3 REQ-M15 — Manual rescue. A mechanically available two-step manual rescue mechanism remains operable under the specified power-loss and hydraulic-failure conditions, does not depend on ECU aliveness, requires deliberate two-step activation and creates no unintended free-float. Interface, access location, actuation force, sequence, indication, lock and capture behaviour, pinch hazards, post-rescue state and re-enable procedure are configuration-controlled in ICD-001. A textual definition does not close the requirement: physical test under the combined power and hydraulic failure is required.

7.4 REQ-M21 — Lateral-g policy (temporary until SIM-001, HIL and four-corner evidence).

Element	Rule
SPORT→COMFORT inhibit	Immediate when ay exceeds PARAM-M21-AY-HIGH, for automatic and manual requests. A driver override may queue a request but cannot defeat the interlock.
Re-enable	Only after ay stays below PARAM-M21-AY-LOW for PARAM-M21-HOLD
Filtering and latency	A validated filter and a measurement-to-inhibit latency budget, both biased to inhibit
Sensor validity and faults	Plausibility and validity criteria for lateral acceleration, yaw rate, roll rate and relevant macro and lock positions; degraded operation, inhibit, disagreement thresholds, stale-data handling and recovery rules; a single lateral-acceleration sensor cannot alone trigger an unsafe transition (REQ-M27)
Transients	Noise handling; yaw and roll transients; braking and combined manoeuvres
COMFORT→SPORT	Also conditioned on lateral acceleration, yaw rate, roll rate and macro position limits (parameters OPEN under SIM-001 / HIL), so that seat-belt drive and lock engagement under cornering load cannot cause unconstrained motion or lock damage
Roll gradient	COMFORT steady-state roll gradient at V-L design payload limited by PARAM-M21-ROLL-GRAD
Manual rescue	Separate procedure (§7.3), for emergency and maintenance only

7.5 Progressive-stop curve (REQ-M09a / M09b). STR-001 defines a complete force-displacement and energy-absorption curve, not a single peak value: trigger displacement; initial engagement force; slope (progressive force rise); peak force; absorbed energy; residual stroke; temperature and rate sensitivity; production tolerance; repeatability; post-event inspection and replacement criteria. The stops absorb the design energy (§6.4) independently of Lorentz authority. The energy arithmetic of §6.4 is a sizing illustration, not proof of crash performance; the full load path and stop performance are demonstrated by validated nonlinear analysis and instrumented component and system tests.

7.6 REQ-M25 — Horizontal load path. Closed before any integration-ready claim. Define and verify: longitudinal and lateral stiffness; yaw restraint; free play; load sharing between capture frame, links, stops and SPORT locks; braking, lateral, combined vertical / horizontal, crash, and propulsion / brake torque-reaction loads; asymmetric four-corner loading; connection strength, fatigue, local buckling, fastener slip and failure containment. The Lorentz stage is never credited as the primary crash or horizontal path. The order-of-magnitude loads of §6.3 and §11.4 are not certification evidence; structural calculation and test correlation establish the final design loads and margins.

7.7 REQ-M27 — Sensor functional independence. Redundancy means functional independence, not duplication. Common-cause analysis covers: common power supply; ECU; communication bus; software; mounting; environmental exposure; calibration error; vibration-induced failure. Independence review also covers shared signal processing. A safety monitor independent of the primary control path is to be evaluated by SAF-001 for critical mode arbitration and force-bound enforcement; it is not assumed to exist until it is designed and tested. No single sensor failure creates an unsafe mode transition.

7.8 REQ-M28 — Fault classes and recovery from S3. ICD-001 and SAF-001 map every fault code to exactly one class, define the reset authority and verify the transitions by HIL and fault injection.

Class	Fault type	Allowed recovery
A	Recoverable transient fault	S3→S2 only after the fault-specific diagnostic-clear criteria, dwell time (PARAM-FAULT-DWELL), sensor plausibility checks and all relevant interlocks pass
B	Latched safety fault	S3→S1 only after authorised reset and recorded fault review
C	Critical capture, structural, lock-integrity or unresolved sensor-common-cause fault	No automatic return to S2 or S2S; the system stays captured in S3 (or S4 on power loss) until safe manual service intervention, entered through S8 when stationary

S3→S2 is prohibited while any safety-relevant fault remains active, stale, intermittent or unclassified. If S3 was entered from S2S the locks are still engaged; releasing them on the way to S2 is permitted only when the class A conditions and the §7.4 lateral-g policy are both satisfied. If S3 was entered from S1 (self-test or precharge fault), recovery is to S1 only and never directly to S2. The §7.4 interlock is an additional interlock and not a complete fault-recovery policy.

8. State machine (S0-S8)

State names are authoritative. SPORT corresponds to S2S; COMFORT to S2.

8.1 States.

State	Name	Rule
S0	Safe / captured / power-off	No powered release; capture remains (passive condition equivalent to S4)
S1	Precharge / self-test	Capture OK; 48 V path checked
S2	COMFORT isolation	Full envelope; capture mandatory
S2S	SPORT locked	All required locks confirmed; stages held
S3	Degraded	Reduced authority; force bounded
S4	Power-loss capture	Passive support; valves fail-closed; envelope per §7.2
S5	Impact preparation	Prepare stops; inhibit high-rate unlock
S6	Impact / arrest	Stops active; no magnet-energy dependence
S7	Post-event safe	No automatic re-enable
S8	Maintenance lockout	Physical lockout engaged

8.2 Transition table.

From	To	Condition / authority
S0	S1	Power-on self-test pass
S1	S0	Controlled power-off command
S2 / S2S	S0	Controlled power-off command, vehicle stationary; capture retained
S1	S3	Self-test fail or critical fault
S1	S2	Driver or automatic COMFORT selection
S2	S2S	SPORT request; REQ-M21 conditions met; all required locks report engaged
S2S	S2	Release permitted by REQ-M21; all locks report released
S2	S3	Critical fault, sensor invalidity, thermal derate, lock fault or accumulator pre-charge loss (fault class per ICD-001)
S2S	S3	Same causes; this transition shall not release locks
S3	S2	Class A faults only (§7.8); not when S3 was entered from S1; locks release only if the §7.4 policy also permits
S3	S1	Class B faults: authorised reset and recorded fault review (§7.8)
S3	S4	Power loss
S2 / S2S	S5	Impact-preparation trigger: pre-crash sensor class and/or deceleration threshold (OPEN until ICD freeze)
S5	S6	Measured impact / stop contact
S5	S2 / S2S	Impact-preparation abort
S5	S4	Power loss
S6	S7	Event handling complete
S7	S1	Manual clear by authorised personnel only
S1 / S2 / S2S / S3 / S7	S8	Physical lockout engaged; only when stationary
S8	S1	Lockout removed by physical key / procedure
S4	S1	Power restored and self-test pass (only exit from S4)
Any powered state	S4	Power loss; valves fail-closed; engaged locks spring-applied fail-locked. Power loss in S0 causes no state change.

Class C faults have no automatic transition out of S3 (§7.8). Verification of every transition is OPEN (open item 9).

9. Safety analysis (FMEA and preliminary HARA)

Ratings are PRELIMINARY screening labels for planning, not an ISO 26262 ASIL assignment. SAF-001 at G2 delivers the system boundary, HARA, safety goals, functional and technical safety concepts as applicable, FMEA, FTA, common-cause and dependent-failure analysis, single-point and latent-fault analysis, diagnostics and coverage rationale, fault-injection plans, the safe-state argument, traceability and independent review. The applicability of standards is decided for the actual vehicle and module integration. No ASIL, ISO 26262 compliance, crash certification or completed safety case is claimed. Scales: S severity, O occurrence, D detection difficulty, each 1–10; $RPN = S \times O \times D$ is a screening rank only. Mechanical capture, passive support and progressive stops are independent of E/E and form the primary safety backbone.

9.1 Summary FMEA.

Failure mode	Effect	Mitigation	S	O	D	RP N
Carriage jam	Loss of micro authority	Sensors; stops; passive support	9	3	4	108
Link fracture	Local free-motion risk	Capture frame; stops	9	2	5	90
Leveling stuck	Stroke consumed	Level sensor; S3	8	3	3	72
Hose burst	Loss of support / leveling	Pressure sense; stops; isolation valves at the cylinder port (§7.2)	9	3	3	81
Coil overheat	Derate / shutdown	Thermal model	7	3	3	63
Magnet Km loss	Force-constant drop	Temperature limits; derate	7	2	4	56
SPORT lock fails to engage	Remains COMFORT	Status sensors; driver alert	6	3	2	36
Partial lock (1-2 of 4)	Asymmetric stiffness	Inhibit S2S declaration	8	2	2	32
False COMFORT at high lateral g	Excess roll	REQ-M21 inhibit	7	2	3	42
Horizontal path overload	Capture shear exceedance	REQ-M25; structural stops	9	2	4	72
Software / OTA fault	Bad command	Bounds; S3 / S4	8	3	4	96
Unintended SPORT lock at speed on rough road	Handling / comfort conflict	Arbitration inhibit; alert; R-17	8	3	3	72
Accumulator pre-charge loss	Macro response degraded	Pressure sense; S3; R-18	7	3	3	63
Hydraulic leak / fire	Loss of support; fire risk	Fluid selection; containment; R-19	9	2	4	72
Common-cause sensor fault	False mode or force	Functionally independent sensing; independent safety monitor to be evaluated (REQ-M27); R-20	8	3	3	72
Magnet demagnetisation	Force-constant drop	Temperature limits; derate; R-21	7	2	4	56
Power loss	Active force lost	S4 with §7.2 limits (ratings in SAF-001)	9	—	—	n/a

9.2 Preliminary HARA (screening). S/E/C are candidate severity / exposure / controllability classes, distinct from FMEA S.

Hazard	Scenario	S/E/C candidate	Safety goal (draft)
Loss of capture	Carriage or link failure	S3/E3/C3	Maintain capture under single failure
Uncontrolled vertical motion	Force runaway	S2/E3/C2	Bound force; enter S3 or S4
Asymmetric SPORT lock	Partial lock + lateral load	S2/E3/C2	Inhibit S2S unless the lock set is complete
Horizontal overload	Brake / crash without sized path	S3/E2/C3	Size REQ-M25 paths in STR-001
Unexpected height drop	Valve open at speed	S2/E3/C2	Fail-closed valves
Post-crash automatic re-enable	Uncommanded motion after event	S2/E2/C2	S7 holds until manual clear
Unintended SPORT lock	False SPORT engage	S2/E3/C2	Inhibit; alert; R-17
Unlock under high lateral g	Release request in a turn	S2/E3/C2	REQ-M21

Process candidates are listed in Annex A. Vehicle category and driving-licence mass thresholds (for example the 3,500 kg threshold in some jurisdictions versus PARAM-GVW-VL) are assessed in REG-001.

10. Actuator map, gap envelope and thermal

10.1 Km evidence chain. PARAM-KM stays an assumption until validated by measured force-current-power-gap data. The verified ACT-001 map, not nominal Km, is the design input for control and thermal sizing.

Axis	Sweep / points	Record
Air gap	Across the full tolerance envelope, min to max	Force, current, power
Temperature	Cold, nominal, hot	Force, resistance, magnet temperature
Current	RMS region to peak	Force, power, thermal rise
Duty cycle	Continuous and pulsed	Thermal equilibrium
Manufacturing sample	Minimum agreed number of samples	Spread and capability
End-of-life surrogate	If available	Derating note

10.2 Gap tolerance envelope. PARAM-LORENTZ-GAP is a complete tolerance-controlled envelope, not a nominal range. ACT-001 demonstrates force capability and safe control authority across the validated envelope, including the horizontal relative motion allowed by REQ-M25 (REQ-M26). The 2-5 mm gap remains a target envelope until a full tolerance stack (manufacturing, bearing / free play, thermal expansion, dynamic deflection, wear, crash deformation) is closed. No performance claim relies on the assumed Km before test or validated analysis.

Worksheet field	Purpose
Nominal / minimum / maximum gap	Design centre; contact-risk bound; authority bound
Assembly tolerance	Build variation
Structural deflection under load	Operating shift
Thermal growth	Hot / cold shift
Vibration-induced motion	Dynamic envelope
Manufacturing tolerance	Part capability
Sensor error	Control uncertainty
Bearing / free play, wear	Gap growth over life
Crash deformation	Post-event gap
Horizontal free play	REQ-M25 / REQ-M26 coupling
Stop-to-gap relationship	End-of-travel coordination

10.3 Thermal. Continuous thermal design is dominated by copper loss at the RMS operating point; continuous and transient limits are separated. Continuous RMS capability and REQ-M11 stay OPEN until bench evidence closes the thermal model. Demagnetisation (R-21) is evaluated across temperature and duty cycle. SPORT continuous copper demand is expected to be near zero because the isolation stages are held; this is design intent and is not verified (thermal bench evidence OPEN).

11. Mode sequences and lock architecture

11.1 COMFORT→SPORT (time per PARAM-MODE-TIME).

Step	Action	Interlock
1	Mode request accepted by arbitration	No active inhibit; §7.4 COMFORT→SPORT conditions met; no fault
2	Macro / leveling drives upper body to seat band	Stroke sensors in range
3	Seat contact detected at required corners	Contact flags true
4	Positive locks command engage	Actuator enable healthy
5	All required locks report engaged	Incomplete set aborts to COMFORT
6	Isolation stages held; state S2S	Driver indication ON

SPORT is never declared while a required lock is disengaged.

11.2 SPORT→COMFORT. Permitted only per §7.4. When allowed: (1) locks command release; (2) all report released; (3) macro envelope re-enabled; (4) micro control re-enabled; (5) state S2. COMFORT is never declared while a required lock is engaged. Manual rescue (§7.3) is a separate two-step procedure.

11.3 Lock architecture (LCK-001).

Element	Function	Status
Seat faces	Upper hard-point to lower receiver at the seat band	OPEN — geometry in ICD-001
Positive lock	Pin, claw or hydraulic clamp preventing separation; per-corner status required	BASELINE
Actuation	Hydraulic or electromechanical; may share macro power	STUDY
Fail behaviour	Fail to a known state with driver alert; never free-float	BASELINE
Crash role	May share load; progressive stops remain the primary energy path	BASELINE

11.4 Horizontal load path summary (Class L; maximum-study basis is the sizing basis).

Load	Order of magnitude (2,650 kg → 3,150 kg)	Path
Brake 1 g	≈26 kN → ≈31 kN total	Capture shear + links + SPORT lock if engaged
Lateral 1 g	≈26 kN → ≈31 kN total	Same; roll couples into the vertical towers
Crash 20 g horizontal class	≈520 kN → ≈618 kN total	Structure + horizontal stops; not Lorentz
Vertical crash with ×2 asymmetry	≈260 kN → ≈309 kN per tower	Stops + cell; R-08

12. Verification framework

12.1 SIM-001 specification.

Item	Planning definition
Model class	Quarter-car / half-car: upper mass, macro k and c, optional micro force, SPORT as high-k constraint
Reference passive	Frozen passive hydro-pneumatic k and c, stated on every comparative plot
Road class	ISO 8608 segments; at least one rough and one smooth profile
Mass states	Empty / occupied / maximum as §6
Metrics	aw (0.5–80 Hz), VDV, MSDV (0.1–0.5 Hz) per ISO 2631-1; axes, weightings (Wk, Wd; Wf for MSDV) and measurement point (seat surface or floor) stated per MCRS-TP-001
COMFORT policy	§7.4 lateral-g policy and PARAM-M21-ROLL-GRAD
Outputs	Transmissibility; band RMS; stroke histogram; roll proxy SPORT vs COMFORT; transition time history
SIM-C1	COMFORT lower RMS in at least one agreed band than the frozen passive on a rough profile
SIM-C2	SPORT higher roll stiffness (lower roll) than COMFORT on the same lateral step
SIM-C3	Macro within its stroke without continuous stop hammering on the agreed rough profile
SIM-C4	No unconstrained free-float interval in any transition case

SIM-C1 to SIM-C4 are TARGET after a valid run. Results are PRELIMINARY until independent review. SIM-C1 is an early target only and cannot support a broad “overall comfort superiority” claim. SIM-001 freezes vehicle and payload, speed, tyre, reference-passive parameters, road profile, frequency bands, filtering and uncertainty; it reports full-band and weighted RMS acceleration, VDV where appropriate, peaks, stroke usage, saturation, handling and attitude response and the trade-offs, compared with a frozen passive reference under repeatable conditions and independent review. Public claims match the actual evidence level.

12.2 Evidence ladder and demonstration mapping.

Level	Scope	Demonstration step	Gate	Status
L0	Architecture and requirements (this document)	D0, D1 (aids, not tests)	G0	BASELINE
L1	SIM-001 numerical run with sensitivity and combined manoeuvres	D2	G1	OPEN
L2	Single-tower bench: force-gap-current, thermal RMS, power loss, lock cycle, stop contact	D3	G2	OPEN
L3	Four-corner / HIL: asymmetry, mode transitions, stops, fault injection	D4	G3	OPEN
L4	Optional mule: ISO 2631-1 and handling on a donor chassis	—	G4	OPTION

Demonstration and evidence share one ladder, so no parallel marketing track can diverge from controlled results.

12.3 Key verification cases.

Requirement	Case set	Level
REQ-M07	Controlled power cut from COMFORT and from SPORT at defined payload and temperature; record detection time, valve closure time, peak displacement and velocity, final settle, free-float check, passive-support check; brown-out cases developmental	L2, L3
REQ-M15	Two-step rescue under power loss, under hydraulic loss and under both; confirm no free-float and the re-enable procedure	L3
REQ-M21	Threshold inhibit; hysteresis and hold re-enable; single sensor fault; sensor disagreement; yaw / roll transient; braking combined; override attempt; COMFORT→SPORT in a turn	L1, L3
REQ-M25 / M26	Stiffness, free-play and load-share memo; combined-load analysis; gap minimum under maximum horizontal excursion	L2
REQ-M09a / M09b	Force-displacement sample; impact surrogate; post-event inspection (§7.5)	L2, L3
REQ-M27	Common-cause analysis; fault injection on shared power, bus and ECU; independent-monitor trip test	L3

Detailed procedures live in TST-001 and TST-002 (T2).

12.4 Evidence record requirements (every L1-L3 result).

Field	Requirement
Software / parameter set	Version identifier and configuration hash
Hardware configuration	Tower serial or surrogate ID; lock fitment; sensor set
Road / profile ID	ISO 8608 class or measured profile reference
Payload / mass state	Empty / occupied / maximum as §6
Mode	COMFORT (S2), SPORT (S2S) or transition case
Reference passive k / c	Frozen values stated on every comparative plot
Time histories	Force, stroke, acceleration, lock status as applicable
Uncertainty	Method stated where numerical claims are made
Review status	PRELIMINARY until independent review disposition

MCRS-TP-001 (reference test protocol, T2) defines vehicle mass, payload, road profile, speed, tyre pressure, passive baseline, COMFORT, SPORT, sensor locations, sampling rate, filtering, metrics, acceptance criteria, uncertainty and configuration hash.

13. Gates, phases, risks, work packages and open items

13.1 Gates G0-G5.

Gate	Intent	Exit criteria (planning)
G0	Concept controlled	Baseline issued; legal owner recorded
G1	Interface and simulation ready	ICD draft including §7.2 limit structure; SIM assumptions frozen; clash check; packaging closed per §4 (signed datum-based drawings, worst-case tolerance stack, independent review; residual ≥ 0 or approved re-baseline); cabin definition and PARAM-FLOOR-HPOINT confirmed; aggregate-P90 mass method and boundaries declared; D2 plot pack v0.1; independent reviewer appointed
G2	Single-tower evidence	Force, thermal, power-loss and lock bench; ACT-001 map started; REQ-M07 limits, REQ-M09a screening curve and REQ-M25 values frozen; ENE-001 worst-case budget reviewed; supplier-supported aggregate P90 mass compliant or approved re-baseline
G3	Four-corner / HIL	Mode transitions; stop-absorption verification; fault injection; REQ-M25 path verified; REQ-M15 trial
G4	Optional mule	ISO 2631-1 and handling on a donor chassis
G5	Licence / transaction readiness	Buyer-accepted package; commercial term sheet; IP filing status; ownership chain

13.2 Phase schedule (planning; no commitment without funding).

Phase	Month	Content
Ph0	0-2	CAD envelope, lock faces, ICD-001 draft; D0 animation recommended; procurement package after ICD baseline
Ph0b	2-6	Supplier lead time (4-8 months planned); COTS surrogate bench; SIM-001 run (D2). Option deliverables in a 90-day window are limited to SIM-001 and ICD-001 draft
Ph1	6-10	Single-tower force, thermal, power loss, lock cycle (D3); ACT-001 map
Ph2	10-15	Four-corner 6-DOF; mode transitions; stop-curve verification; power loss from each mode; REQ-M25 path (D4)
Ph3	15-21	Optional donor-chassis ISO 2631-1 and handling evaluation

13.3 Risk register (likelihood × impact, 1-5).

ID	Risk	L	I	L×I	Owner	Mitigation	Gate
R-01	Core tower mass exceeds the limit at PARAM-MASS-CONFIDENCE	4	4	16	Module	Supplier mass data; structure trade; revise limit or design under CFG-001	G1 / G2
R-02	Km not achieved / not demonstrated by measured map	3	5	15	Actuator	FEA + supplier map; margin on force	G2
R-03	Lorentz thermal or force-gap limits	3	5	15	Actuator	Thermal bench; derate strategy	G2
R-04	Macro-micro clash or carriage failure	3	4	12	Mechanical	Clash check; stops; sensing	G1
R-05	Stroke consumed without leveling	3	4	12	Hydraulics	Leveling authority; S3 fallback	G2
R-06	Regulatory path unclear	3	4	12	Homologation	REG-001 pathway study	G3
R-07	Vehicle P90 exceeds host mass ceiling	4	4	16	Mass	Host WBS reserve; P90 tracking	G1
R-08	Per-tower crash load versus capture frame / stops / hardpoints	4	5	20	Structures	STR-001 curve on maximum-study mass	G2
R-09	Comfort unproven until L1/L2 evidence	4	4	16	Simulation + PoC	SIM-001 (D2); bench (D3)	G1
R-10	EMI / field-exposure cost	2	4	8	EMC	EMC plan (REQ-M12)	G3
R-11	Hydraulic mass above STUDY	3	3	9	Hydraulics	HYD-001 supplier data	G2
R-12	Liability if safety case or claims exceed evidence	2	5	10	Legal + Safety	SAF-001; non-claims; counsel review of wording	G2
R-13	SPORT lock reliability / partial engagement	3	4	12	Lock	Status sensing; S2S inhibit	G2
R-14	Mode-arbitration false positive / negative	3	3	9	Controls	HIL; hysteresis	G3
R-15	Horizontal restraint undersized (REQ-M25)	3	5	15	Structures	STR-001; ICD geometry	G2
R-16	Clearance-stack energy underestimated	3	4	12	Structures	Relative-velocity crash model (\$6.4)	G2
R-17	Unintended SPORT lock at speed on rough road	3	4	12	Systems	Arbitration inhibit; alert	G2
R-18	Accumulator pre-charge loss	3	3	9	Hydraulics	Pressure sensing; S3	G2
R-19	Hydraulic leak / fire	2	5	10	Hydraulics	Fluid selection; containment	G2
R-20	Common-cause sensor fault	3	4	12	Systems	Functional independence; independent safety monitor to be evaluated (REQ-M27)	G2
R-21	Magnet demagnetisation	2	4	8	ACT-001	Temperature limits; derate	G2
R-22	Packaging residual negative at the quoted heights (\$4)	4	4	16	Packaging	Re-baseline decision; closure options \$4	G1
R-23	IP not filed before T1 distribution; ownership chain incomplete	3	5	15	Counsel / IP-001	Provisional filing; disclosure register	Before broad T1
R-24	Single-source or long-lead supplier items (Lorentz, stops, locks)	3	4	12	Module	Dual-source questionnaire (\$15.1)	G1
R-25	Horizontal free play incompatible with the Lorentz gap	3	4	12	Structures / Actuator	REQ-M26; decoupled-carriage option	G1
R-26	Name MCRS conflicts with an existing mark; or ownership of the name is unclear	2	4	8	Counsel / IP-001	Trademark clearance before external use of the name	Before broad T1

13.4 Open Items Register (complete).

#	Open item	Owner path	Target gate
1	Packaging residual to ≥ 0 : re-baseline decision; cabin definition and PARAM-FLOOR-HPOINT confirmed	\$4 / CFG-001	G1
2	REQ-M07 numeric limits ($t_{det,max}$, $t_{valve,max}$, z_{max} , v_{max}) frozen	\$7.2 / SAF-001 / ICD-001	G2
3	REQ-M15 operability under dual failure (power + hydraulic), physical trial	\$7.3	G3
4	REQ-M21 filter, latency, hold time, sensor-disagreement behaviour and COMFORT→SPORT conditions	\$7.4 / SIM-HIL	G2
5	REQ-M25 numeric free play, stiffness and load share	\$7.6 / STR-001 / ICD-001	G2
6	Progressive-stop curve frozen (G2); design energy event verified (G3)	STR-001	G2 / G3
7	IP ownership chain completeness and provisional filing	IP-001 / counsel	Before broad T1
8	Configuration discipline — one value, one place	CFG-001	G1
9	State-machine verification (S3 entry and exit, S4 exit, S5 trigger)	\$8 / HIL	G2
10	Km evidence chain and gap-tolerance envelope	ACT-001	G2
11	48 V total system current budget	ENE-001	G2
12	Aggregate tower P90 mass: method and boundaries declared (G1); supplier-supported result or approved re-baseline (G2)	Supplier data / CFG-001	G1 / G2
13	Horizontal free play versus Lorentz gap (REQ-M26)	STR-001 / ACT-001	G1
14	V-L range target consistency (§14.2)	ENE-001	Before external claim
15	REQ-M27 common-cause analysis and independent-monitor concept	SAF-001	G2
16	Mode-time allocation (lock time, detection and valve times, inhibit exclusion)	ICD-001	G2
17	Trademark clearance and ownership of the name MCRS	Counsel / IP-001	Before broad T1
18	Fault-code to class (A, B, C) mapping and reset authority (REQ-M28)	ICD-001 / SAF-001	G2

13.5 Immediate work packages.

Priority	Work package	Closure deliverable
G1	Freeze CAD envelope, lock faces, leveling, fail-closed valves; packaging re-baseline decision	3-D package + clash report + residual status
G1	ICD-001 draft (mechanical / electrical / communication / safety / lock) with \$7.2 limit structure	Buyer-readable ICD set
G1	SIM-001 assumptions and first COMFORT-vs-passive run (D2)	Plot pack v0.1 + configuration hash
Pre-T1	D0 architecture animation (recommended)	Shared visualisation file
G2	Single-tower force, thermal, power loss, lock cycle (D3)	Bench data package
G2	Formal HARA + FTA; common-cause analysis (REQ-M27)	SAF-001 work products
G2	STR-001 stop curve + REQ-M25 / REQ-M26 horizontal loads	Structures memo
G3	Four-corner rig mode transitions (D4)	6-DOF + transition-time records
G5	Path A Option / term-sheet support	Commercial pack under counsel (COM-001)

13.6 Gate closure matrix (minimum closure evidence).

Work package	Minimum closure evidence	Gate / status
CFG-001 / packaging	3-D datum model; signed tolerance stack; clearance, travel and service envelope; independent review	G1 · OPEN · P0
MASS-001	Supplier-supported mass distributions; declared correlation; aggregate P90 method; compliant result or approved re-baseline	G1 method · G2 result · OPEN
ENE-001	Worst-case power and current budget; transient, thermal, protection, wiring and EMC review	G2 · OPEN
ICD-001 / REQ-M07, M15	Frozen numeric limits, interfaces, rescue procedure and fault matrix; bench and fault-injection evidence	G2 / G3 · OPEN
ACT-001	Force-current-gap-temperature map, demagnetisation and efficiency evidence	G1 / G2 · ASSUMPTION
STR-001 / REQ-M25	Load-path model, stop curve and energy, tolerances, analysis correlation and physical test	G2 / G3 · OPEN
SAF-001	HARA, FMEA, FTA, common-cause analysis, safety requirements, diagnostics, fault injection and independent review	G2 · OPEN
SIM-001	Frozen test conditions, comparative multi-band results, uncertainty and independent disposition	L1 target · D2
IP-001 / counsel	Ownership chain, disclosure register, filing decision and receipt if filed, prior art, claim plan and staged FTO advice	Before T1 / T3 disclosure

14. Host variants (Path B) — engagement targets

Host figures are STUDY for vehicle programmes. Path A does not require the buyer to adopt host powertrain or GVW figures. Host budgets use PARAM-OCC-POLICY. Engine entries are class descriptors, not brands. LOCKED geometry values are engagement targets subject to §4.

14.1 Geometry.

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Overall length	5,000 mm	5,500 mm	LOCKED
Overall width (mirrors excluded)	2,000 mm	2,200 mm	LOCKED
Overall height (SPORT / seated)	1,700 mm	2,000 mm	LOCKED
Wheelbase	3,000 mm	3,300 mm (front axle to mid of rear tandem)	LOCKED
Ground clearance (SPORT)	200 mm	300 mm	LOCKED
Wheels / drive	4×4	6×6 (front axle + dual rear tandem)	LOCKED
Design occupants	4	6	LOCKED
Passenger cell	Full CFRP structural cell	Full CFRP structural cell	BASELINE
Roll structure	FIA-referenced CFRP / hybrid (no homologation claim)	Same	BASELINE
GVW hard ceiling	PARAM-GVW-VL	PARAM-GVW-VH	LOCKED

In every external-facing table the quoted heights read “LOCKED TARGET / PACKAGING GATE OPEN”. Height and ground clearance are subject to the §4 re-baseline decision. Cell mass budget ≤ 280 kg (V-L) and ≤ 420 kg (V-H); roll-structure budget ≤ 45 kg and ≤ 70 kg (TARGET). FIA Appendix J concepts are a philosophy reference only; load cases are OPEN until edition freeze.

14.2 Powertrain (hybrid ICE + electric assist, class descriptors).

Parameter	V-L	V-H	Status
Architecture	Hybrid (ICE + e-drive)	Hybrid (ICE + e-drive)	LOCKED
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	LOCKED
ICE power	≈ 430 –450 kW class	≈ 460 –500 kW class	TARGET
Electric traction	≈ 150 –180 kW total	≈ 200 –250 kW total	TARGET
Battery usable energy	20 kWh	40 kWh	LOCKED
Fuel tank	80 L	150 L	LOCKED
Combined range target	1,000 km	700–800 km	STUDY

Range consistency check. 1,000 km on an 80 L tank requires ≈ 8 L/100 km average (before any battery credit) for a ≈ 3.4 t vehicle with a 4.0 L twin-turbo V8-class engine; this is aggressive, especially at 120 km/h. V-H (150 L over 700–800 km) implies ≈ 18.8 –21.4 L/100 km. Both remain STUDY pending road-load calculation at 90 / 120 km/h under ENE-001; the V-L figure is recomputed before any external claim. Implied battery specific energy is ≈ 138 Wh/kg (V-L) and ≈ 154 Wh/kg (V-H), to be confirmed against supplier pack data. These are arithmetic implications of the stated targets, not verified consumption results; road-load and duty-cycle models are reviewed before any external claim.

14.3 Closed mass budgets (STUDY).

WBS line	V-L kg	V-H kg
CFRP passenger cell	280	420
Roll members	45	70
Closures, glass, panels / reinforced exterior	230	400
Interior, HVAC, cabin LV	200	330
Lower chassis	340	720
ICE and ancillaries	275	390
Hybrid transmission / coupling	160	190
E-motors and inverters	130	180
Battery (usable energy per §14.2)	145	260
HV/LV and cooling	95	160
Axles, steering / three-axle 6x6 driveline	260	580
Wheels, tyres, brakes	210	480
Isolation, four corners (Class S / Class L basis)	280	380
Optional light protection package	—	200
Fuel tank and reference fuel	75	140
Exhaust and fluids	70	100
Programme reserve	70	150
Curb mass, no occupants	2,865	5,150
Occupants (PARAM-OCC-POLICY × 4 / × 6)	400	600
Cargo / mission payload	100	400
GVW design case	3,365	6,150
Margin to ceiling	635	850

V-H includes the optional 200 kg protection package; if omitted it moves to reserve.

Traceability. V-L occupied upper body $\approx 1,255$ kg (cell, roll members, closures, interior, occupants, luggage) $\rightarrow \approx 314$ kg per corner, inside the Class S basis of 310–350 kg. V-H occupied upper body $\approx 2,220$ kg ($\approx 2,420$ kg with the protection package) $\rightarrow \approx 555$ –605 kg per corner, above the Class M basis and below the Class L reference (662.5 kg); the isolation line therefore uses the Class L basis.

14.4 Host requirement additions.

ID	Requirement	Status
REQ-V01	V-L GVW claims shall remain within PARAM-GVW-VL including the stated occupant policy	TARGET
REQ-V02	V-H GVW claims shall remain within PARAM-GVW-VH including the stated occupant / payload policy	TARGET
REQ-V03	Both variants implement a CFRP passenger cell within the §14.3 mass budgets	TARGET
REQ-V04	Both variants implement an FIA-referenced roll structure; no homologation claim without a formal process	BASELINE
REQ-V05	V-L ICE: V8, 4.0 L twin-turbo class	LOCKED
REQ-V06	V-H ICE: V12, 6.0 L twin-turbo class	LOCKED
REQ-V07	Range: V-L 1,000 km; V-H 700–800 km, both pending road load at 90 / 120 km/h	STUDY

15. Supplier and ICD outline

15.1 Supplier questionnaire themes (no supplier is committed; catalogue values are not programme evidence).

Domain	Questions the supplier must answer
Hydraulic macro	Cylinder force at pressure; seal life; fail-closed valve response time; valve location relative to the cylinder port; orifice-hold capability
Lorentz / actuator	Km measured across gap tolerance; continuous and peak force; thermal map; mass P50 / P90 with process capability; demagnetisation limits; tolerance to horizontal relative motion
Lock	Engage / release time under load; status sensing; shear capacity; contamination tolerance
Stops	Force-displacement curve; energy rating; temperature sensitivity; inspection interval
Sensors	Stroke, force, pressure and temperature accuracy; fault behaviour; independence from control sensors
Electronics	48 V drive capability; current limit; diagnostic coverage; EMC class
Commercial	Single-source status; lead time; second-source option; change-control and IP-clause willingness; quality system and traceability level

15.2 ICD-001 minimum chapters. Hardpoint geometry and loads; hydraulic ports and fluid specification; electrical power and signals (48 V class with the full current envelope); mode and interlock logic summary; REQ-M07 limits; progressive-stop curve

reference; thermal limits; installation torque and sequence; maintenance and rescue release; service access; EMC and grounding; configuration identity and change control.

15.3 Configuration management. All TARGET and STUDY values that become design inputs are configuration-managed under CFG-001. This baseline remains the narrative reference; freeze occurs in the ICD and part drawings, not by revision of this document alone. One value, one place: a frozen value appears in exactly one authoritative location (§3 or a named downstream document); narrative sections cite the parameter ID and do not restate a conflicting number. Conflicting numbers are defects corrected at the next revision.

CFG-001 item	Required content
Document	Revision ID, date, supersedes
Parameter change	Old value, new value, reason
SUPERSEDED mark	Prior value retained for traceability
OPEN-item closure	Evidence reference and gate
Review finding	Disposition and owner
Evidence configuration hash	Software, parameters, hardware serial class
Disclosure tier of release	T0-T3
Change request	Impact analysis, affected requirements and artefacts, approver, date, revision and re-verification scope

16. Transaction readiness (Paths A, B, C)

16.1 Path A (module licence) — primary commercial route. Field licence or supply of MCRS-MOD-001 without requiring the buyer to adopt host powertrain or GVW figures. Commercial terms (Option validity, exclusivity, fees, credit) are counsel-controlled T3 content in COM-001. T1 retains only this reference: a Path A Option is available under COM-001; non-exclusive evaluation by default; limited exclusivity as a paid add-on.

16.2 Path B (vehicle programme). Uses the hosts of §14 when the buyer funds a vehicle path. Not required for Path A.

16.3 Path C (IP assignment). Price and conditions are bilateral, subject to counsel, patent filing status and evidence gates. Contracting party: Amirbahador Atai (individual) until formal assignment to a legal entity.

16.4 Boundary statement. This document is not a certificate, type approval, warranty, patent grant or price list, and does not constitute an offer of exclusive sale. Contradicting external presentations shall be withdrawn.

16.5 Intellectual property filing status. Per project record, no patent application had been filed as of 2026-10-07. The Legal Owner confirms this status before each release of this document. When a provisional application is filed this section is updated to: Provisional application No. [number], [office], [date]. Recommended internal rule (counsel to confirm): file a provisional application before further broad T1 distribution to third parties; an NDA alone may be insufficient in absolute-novelty jurisdictions. Any earlier file that contained T3 content shall be confirmed as not distributed externally. Before any “patent pending” statement a filing receipt and counsel-approved wording are required. A prior-art search is not an FTO opinion, and a filing is not proof of novelty, validity or commercial exclusivity. IP-001 contains invention disclosures, named inventors, dated contribution records, employer and contractor assignments, chain of title, disclosure and NDA history, prior-art and patent-landscape search, claim strategy, filing jurisdiction and timing, trade-secret controls and a staged FTO review by qualified counsel. No novelty, enforceability or freedom-to-operate conclusion is asserted. The patent-family schedule, invention-disclosure candidates, filing and ownership record and term-sheet topics are T3 content reserved for MCRS-DOC-00-T3A (to be issued under counsel) and are not part of this file.

16.6 Conditions precedent to any assignment or exclusive licence.

#	Condition	Owner
1	Provisional or full filing record in force	Counsel / IP-001
2	Invention disclosure and inventor chronology complete	IP-001
3	Prior-art and landscape review recorded	Counsel
4	Ownership chain and assignment path defined	Counsel / COM-001
5	NDA and disclosure-history register current	IP-001
6	Evidence gates agreed (for example ICD + SIM or bench plan)	Technical + commercial
7	FTO opinion path defined (may be a post-signing condition)	Counsel
8	Entity assignment path defined if the seller remains an individual; liability allocation in COM-001	IP-001 / COM-001
9	Trademark clearance for the name MCRS	Counsel

16.7 What Path A delivers (planning summary — full terms in COM-001).

Typically included under a closed Path A licence	Not included unless separately agreed
Rights to use the defined CMFSB module architecture in a named field of use	Worldwide assignment of all related IP
Controlled package: ICD, datasheet, safety behaviour, dual-mode rules	Turnkey certified vehicle or type approval
Agreed engineering support hours	Unlimited R and D for the buyer's entire vehicle line
Updates within a defined revision window if contracted	Guaranteed comfort percentage without L1/L2 evidence

In diligence decks the §16.7 boundary is shown as a single table without additional promises. Prices, exclusivity fees and Option duration remain in COM-001 only.

16.8 Readiness matrix for external engagement.

Use case	Suitable on this baseline?	Condition
NDA + T1 technical diligence	Yes	Under NDA, subject to the T1-release precondition (Annex D.3)
Paid Option negotiation	Yes	Bilateral term sheet; fee not published here (COM-001)
Module licence term sheet	Conditional	ICD draft + SIM or bench plan committed
Large upfront assignment / exclusive sale	No — not on this document alone	IP filing status, FTO counsel, evidence
Production / type-approval claim	No	Explicit non-claim
Advertising measured ride comfort	No	Until L1/L2 configuration-managed results

17. Governance and sign-off**17.1 Master Document Register (single source of IDs).**

ID	Title / scope	Status
DOC-00	This baseline	R5.29 (single layer; R5.26 overlay withdrawn)
DOC-00-T3A	IP schedule (T3, counsel; own revision control)	To be issued separately under counsel
MOD-001	Module datasheet (1–3 page buyer summary; T0-compatible)	To issue from DOC-00
ICD-001	Interface control document	OPEN — draft under Option
SIM-001	Multi-band simulation	OPEN
STR-001	Structural / progressive-stop / crash path	OPEN
LCK-001	Lock system	OPEN
ACT-001	Actuator thermal, force, efficiency, gap map	OPEN
HYD-001	Hydraulic macro stage, valves, leveling	OPEN
SAF-001	HARA / FMEA / FTA / safety case	OPEN
TST-001/002	L2 / L3 test plans	OPEN
MCRS-TP-001	Reference test protocol (T2)	OPEN
CFG-001	Configuration management	OPEN
ENE-001	Energy / power budget detail	OPEN
MASS-001	Aggregate mass analysis: supplier distributions, correlation assumptions, aggregate P90 method (R-01)	OPEN
BRK-001	Brake / horizontal load interface	OPEN
REG-001	Regulatory pathway	OPEN
IP-001	IP / patent filing / trademark clearance (counsel)	OPEN
COM-001	Commercial annex (T3 / counsel)	OPEN

17.2 Independent reviewer challenge list (before G1 credit).

#	Challenge topic
1	Capture claims versus free-levitation misunderstanding
2	Mass at the declared confidence level versus the P90 line-sum and RSS estimate
3	Km assumption exposure in external materials
4	Static floor rule and mass-basis consistency
5	Packaging residual versus LOCKED language
6	REQ-M07 measurability (limits frozen?)
7	REQ-M15 operability under dual failure
8	REQ-M21 filter, single-sensor and disagreement behaviour
9	REQ-M25 / M26 openness versus any integration-ready language
10	Progressive-stop curve completeness and design energy event
11	IP ownership chain completeness
12	Configuration discipline and one-value-one-place
13	State-machine completeness (S3 entry and exit, S4 exit)
14	REQ-M27 independence of the safety monitor

17.3 Priority before serious OEM or investor engagement. (1) D0 animation and ACT-001 / STR-001 / SIM-001 evidence path; (2) SAF-001 including quantitative power-loss limits, sensor independence and horizontal load path; (3) IP-001 filing, prior art, claim chart, ownership chain and name clearance; (4) packaging re-baseline decision and non-negative residual under configuration control.

17.4 Release and sign-off.

Role	Name / status	Date / target
Legal Owner / Author	Amirbahador Atai	2026-10-07
Independent Technical Reviewer	TBD — appointed by PARAM-REVIEWER-DEADLINE	Earlier of 2026-12-05 and G1
Approval Authority	TBD after independent review	After G1 evidence
Documentation authority	MCRS Project Engineering Documentation Authority	2026-10-07

Status: CONTROLLED / Contract Diligence Baseline. Not approved for road use, production or certification credit. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release.

17.5 Release checklist.

#	Check before every external release
1	Every exported copy identifies R5.29 as the controlling issue; earlier issues are archived unchanged for traceability
2	Negative packaging residuals and the P90 mass exceedance are prominent in every external summary
3	All OPEN requirements remain visible in the register and gate tracker
4	IP counsel review is complete before non-NDA disclosure; no unsupported patent or FTO statement
5	Independent engineering review is obtained before any safety or performance credit
6	Patent filing status (§16.5) confirmed by the Legal Owner

Annex A — References, glossary and abbreviations

Reference	Use	Status
ISO 26262	Functional-safety process for E/E elements	Candidate; compliance not claimed
ISO 21448	Safety of the intended functionality (SOTIF)	Candidate
ISO 2631-1	Whole-body vibration evaluation	Reference
ISO 8608	Road-surface profiles for simulation	Reference
ISO 4413	Hydraulic fluid power — general rules and safety (REQ-M24)	Reference
ISO 13849-1	Lock and valve safety functions if used	Candidate
ISO 21780	48 V supply-voltage electrical requirements	Candidate
UN R100 / ISO 6469-3	HV safety if applicable	Candidate
UN R94 / R95 / R14 / R16	Crash and restraint matters if applicable	Candidate
EN 1789 / EN 1865	Ambulance planning references (RA-1 only; no clinical claim)	Candidate
FIA Appendix J	Roll-structure philosophy only (verify article applicability)	Reference

Glossary. **MCRS** — Modular Captured Ride System (product and programme name). **CMFSB** — Captured Mechanical Floating Safety Body (architecture); “floating” denotes compliant isolation inside a bounded envelope and never free-float or levitation. **ATFSV** — retired prefix of earlier document identifiers. **S4** — power-loss passive-support state. **COMFORT / SPORT** — dual modes (S2 / S2S).

RA-1 — reference application: ambulance / clinical-transport planning context (no clinical outcome claimed). **Path A / B / C** — module licence, vehicle programme, IP assignment. **D0-D4** — demonstration ladder. **VDV** — vibration dose value. **MSDV** — motion sickness dose value. **P50 / P90** — median and 90th-percentile planning values. **RSS** — root-sum-square. **RPN** — screening rank $S \times O \times D$. **LOCKED** — programme engagement decision; not packaging closure and not test verification.

Abbreviations. ACT actuator; CFG configuration; COM commercial; ECU electronic control unit; ENE energy; FEA finite-element analysis; FTA fault-tree analysis; FTO freedom to operate; HARA hazard analysis and risk assessment; HIL hardware-in-the-loop; ICD interface control document; ICE internal-combustion engine; LCK lock; PM+EM permanent-magnet plus electromagnet; REG regulatory; SAF safety; SIM simulation; STR structures; TRL technology readiness level; WBS work breakdown structure.

Annex B — Controlled figure descriptions

Figure	Title	Minimum content	Owner
Fig. 1	System context	Upper body, four towers, host hardpoints, layer stack top → bottom	Systems
Fig. 2	Corner force path	Macro, Lorentz, progressive stop, SPORT lock; vertical and horizontal load arrows	Structures
Fig. 3	Mode state graph	S0-S8 with the §8.2 transitions; REQ-M21 inhibit; power loss to S4	Systems
Fig. 4	Power-loss timing	Command loss → detection → valve close → settle inside the §7.2 envelope; lock fail-locked if engaged	Safety
Fig. 5	Packaging stack (V-L)	Overall height, ground clearance, chassis, tower, travel and free play, floor-to-H-point, cabin, roof, residual call-out	Packaging
Fig. 6	Lock seat band	Seat faces, pin / claw class, status sensors, clearances in both modes	LCK-001
Fig. 7	Three-state product diagram	COMFORT / SPORT / S4 side by side with one-line architecture mapping	Systems / MOD-001
Fig. 8	Horizontal free play versus Lorentz gap	Free-play chain, gap minimum, decoupling option (REQ-M26)	Structures / ACT-001

Clearance and stroke stack (nominal → stop). Nominal height is the leveling set-point under stated payload (it shifts with payload if leveling fails). Macro envelope is ± 50 mm about nominal. Free play is PARAM-FREEPLAY-V. Stop stroke is the energy-absorption displacement (STR-001; curve OPEN, not a compliance claim). S5 may reduce effective free play; energy figures are recomputed for the frozen S5 geometry.

Annex C — Acceptance criteria, communication rules, buyer checklist and integration notes

C.1 Path A Option deliverable acceptance (90-day window — planning).

Deliverable	Acceptance criterion (planning)	Not accepted if
ICD-001 draft	Buyer-readable chapters covering hardpoints, power, signals, mode logic, §7.2 limit structure, stops reference	Safety or lock chapters missing
SIM-001 run pack	COMFORT vs frozen passive on at least one rough ISO 8608 profile; configuration hash recorded	No passive reference or no hash
Configuration index	Document IDs and revision status for all artefacts in the Option pack	Conflicting numbers across files

Option acceptance does not imply production readiness, comfort percentage or certification credit.

C.2 External communication rules.

Allowed	Not allowed
Describe architecture: captured body, COMFORT / SPORT, macro / micro, progressive stops	Invent measured comfort percentages
Use the product line and teaser metrics (§1.4, §1.4.1) with status labels	Claim type approval, ASIL or crash certification
Cite TARGET / STUDY / ASSUMPTION labels with their vocabulary	Present STUDY or ASSUMPTION as verified performance
Reference this baseline under NDA for diligence	Publish fixed licence prices
State patent filing status exactly as in §16.5	Imply novelty, grant or FTO; say “patent pending” before a filing number exists
Show the Path A include / exclude table (§16.7) without extra promises	Promise attitude / lean features not in this baseline
Quote core tower mass only with both P50 and P90 and the declared confidence level	Quote tower mass without basis, quote only P50, or quote full-corner mass as module mass
State the packaging residual as in §4	Describe the quoted overall heights as feasible or closed
Describe this issue as “Contract Diligence Baseline — TRL 2-3; open verification gates remain”	Call any issue “final” in a way that implies package closure, verified performance, safety approval or IP protection

C.3 Buyer diligence checklist.

#	Question for the selling party	Where answered
1	What is locked versus still STUDY / OPEN?	§1.7; §3; §13.4
2	Is packaging closed for the quoted overall height?	§4 — No; residual explicit
3	What is the patent filing status?	§16.5 — none on record as of the issue date; the Legal Owner confirms before each release
4	What does Path A include and exclude?	§16.1; §16.7
5	What evidence exists at L1 / L2 today?	§12.2 — currently L0 architecture
6	Who is the contracting party?	Cover; §16.3 — Amirbahador Atai (individual)
7	How are commercial terms controlled?	COM-001 (T3); not published in T1
8	What is the product in one line?	§1.4
9	What is the lowest-cost next demonstration?	§1.8 — D0 / D2
10	Where does the tower mass stand statistically?	§6.1; R-01 — P50 48 kg; P90 line-sum 67 kg; RSS ≈59 kg; limit 55 kg at P90 not met
11	Are quantitative safety limits defined?	§7.2, §7.4 — structure defined; values OPEN

C.4 Integration notes for 4×4 and 6×6 hosts.

Topic	4×4	6×6
Module interface	Four towers at upper-body hardpoints	Same four towers if load class held
Integration risk	Lower — preferred pilot path	Higher (axle, steer, packaging, ground clearance)
First-contract bias	Road / limited-series pilot	Second SKU or buyer-funded utility path
Differentiation	Road / limited series / ambulance pilot	Extreme / specialty / defence-light

Module hardpoints match host chassis stiffness and packaging. Brake and propulsion torque reactions shall not overload horizontal links (REQ-M25). Hybrid powertrain packaging is a host responsibility; the module ICD defines only interface loads, power and signals. Numbers from withdrawn files shall not be mixed with R5.29 parameters.

Annex D — Module datasheet outline (MOD-001) and engagement sequence

D.1 Purpose. MOD-001 is the short buyer-facing datasheet issued from this baseline. It is T0-compatible, introduces no numbers absent from §1.4.1 and §3, and carries no claims absent from §1.2.

D.2 Recommended MOD-001 structure (1-3 pages).

Block	Source in this baseline	Rule
Title + product line	Cover / §1.4	Four clauses only; no extra adjectives
Three-state table	§1.4	COMFORT / SPORT / S4 with architecture mapping
Teaser metrics	§1.4.1	Capture, dual mode, mode-time target class, fail-closed passive support; no mass figure
Path A includes / excludes	§16.7	Single table; no prices
Evidence status	§12.2; §1.8	State the current level (today: L0)
Contact / classification	Cover	Legal Owner name; NDA for further material
Non-claims pointer	§1.2	One line: full non-claims live in this baseline

D.3 Recommended engagement sequence (no new technical complexity).

Step	Action	Document / artefact
1	First contact / teaser	T0: product line + teaser metrics only
2	T1-release precondition: provisional filing completed, or written counsel clearance for disclosure under NDA (recommendation; counsel to confirm); name clearance for MCRS	IP-001 / counsel
3	NDA signed; release this baseline under T1 (without T3 content)	Standard NDA; DOC-00 R5.29
4	Diligence call	Walk §1.4, §4 residual, §16.5 patent status, §16.7
5	Optional D0 animation	Shared visualisation before or with Option
6	Path A Option	COM-001 terms; deliverables SIM-001 + ICD-001 draft
7	Milestones D2 → D3 → D4	Under configuration control
8	Licence or Path C	Only after agreed evidence gates and counsel checks

D.4 Items kept out of MOD-001 and first-contact decks.

Keep out	Reason
Full mass WBS and host GVW tables	Path B detail; distracts from the Path A module story
Km assumption and copper-peak derivation	T1 §3 / T2 ACT-001 material
FMEA score matrix	Screening only; SAF-001 owns formal work
Tower mass figures	Require P50 / P90 basis; T1 only
Attitude / lean / pyramid concepts	Not in this baseline
Any price, royalty or exclusivity fee	COM-001 only
“Patent pending” before a filing number exists	Honest posture per §16.5

Annex E — Comparison to conventional approaches (qualitative only)

This annex places CMFSB in a known design space using architecture language only. It claims no measured superiority, cost advantage or certification. Any quantitative comparison uses SIM-001 or bench evidence under configuration control and restates the §1.2 non-claims.

Approach	Typical behaviour	CMFSB contrast (architecture only)
Soft cabin mounts only	Isolation with limited handling stiffness under lateral load	Adds a SPORT positive lock while retaining capture
Active body control (full vehicle)	Vehicle-level actuators and complex control	Module-scoped four-tower interface; host keeps propulsion and brakes
Magnetic levitation concepts	Free or lightly constrained vertical motion	Not used — body always captured
Passive hydro-pneumatic only	Fixed comfort / handling compromise	Selectable COMFORT vs SPORT without unconstrained free-float
Seat-only isolation	Occupant isolation; cabin structure still harsh	Isolates the structural upper body, not only the seat

One-sentence diligence summary. CMFSB is a mechanically captured, dual-mode (COMFORT / SPORT) isolation module with fail-closed passive support on power loss, offered primarily as a Path A licence object, with packaging residual and actuator evidence still OPEN at TRL 2–3.

Annex F — Master position statement and document map

F.1 Master position statement. R5.29 is the Contract Diligence Baseline for MCRS-MOD-001. It defines a mechanically captured macro-micro isolation architecture with COMFORT and SPORT modes; requirements REQ-M01 to REQ-M28 with verification methods and quantitative-limit structures for power loss, rescue, lateral-g policy, fault-recovery classes, stop curve, horizontal load path and sensor independence; the SIM-001 verification skeleton; gates G0–G5; phases Ph0–Ph3; risks R-01 to R-26; a module-visible mass table with P50 and P90 reported against the declared confidence level; host WBS budgets for V-L and V-H under a 100 kg occupant policy; and an explicit packaging residual (V-L –370 mm and V-H –240 mm on the H-point definition, OPEN, P0). TRL remains 2–3. The Legal Owner is Amirbahador Atai. No test result, certification, patentability or freedom-to-operate is asserted. Remaining work is evidence and formalisation: D0 animation, SIM-001, single-tower bench, ICD-001 and STR-001 freeze, packaging re-baseline, independent reviewer appointment, name clearance and counsel-led IP filing.

F.2 Document map.

Section / Annex	Primary content
Document at a glance	One-page informative summary
§0	Revision control; naming
§1	Purpose, non-claims, value, product framing, KPIs, tiers, vocabulary, demonstration ladder
§2	Architecture, load paths, safety invariants, modes, hold, horizontal restraint
§3	Master Parameter Register
§4	Packaging feasibility gate
§5	Power and current envelope
§6	Mass, static floors, load cases, stop energy
§7	REQ-M01 to REQ-M28 and expanded requirement text
§8	State machine S0-S8
§9	FMEA and preliminary HARA
§10	Actuator map, gap, thermal
§11	Mode sequences, lock architecture, horizontal paths
§12	Verification framework
§13	Gates, phases, risks, open items, work packages
§14	Host variants V-L / V-H
§15	Supplier and ICD outline; configuration management
§16	Transaction readiness; IP status
§17	Governance, register, reviewer challenge list, sign-off, release checklist
Annex A-F	References and glossary; figures; acceptance and communication rules; MOD-001 and engagement; comparison; position and map
Separate file	MCRS-DOC-00-T3A — IP schedule (T3, counsel-controlled)

END OF MCRS-DOC-00 — REVISION 5.29. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release. This is the only end marker of this file.

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